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AA216 : Flight Mechanics and Classical Control

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Problem Statement: Altitude Control with Jerk Stabilization and Variable Mass

Aerospace vehicles lose mass as fuel burns, continuously shifting system dynamics and making fixed-gain controllers unreliable. Standard altitude control systems track only position and velocity, ignoring jerk. This project aims to design a classical control system that tracks a fixed vertical altitude profile for a variable-mass vehicle while stabilizing jerk (the rate of change of acceleration), using Laplace transforms and Dirac delta-based impulse modeling. This project investigates a classical control approach to:

1. Model a variable-mass vertical aerospace system with time-varying dynamics.
2. Track a pre-defined fixed altitude trajectory despite continuous mass reduction.
3. Stabilize jerk using impulse-based modeling via Dirac delta functions.
4. Solve system dynamics analytically using Laplace transforms.
5. Design a classical controller to compensate for variable plant parameters.
6. Analyze closed-loop stability across the full range of operating mass values.
7. Evaluate system response in terms of altitude accuracy and jerk minimization.

Control Objective

Control Objectives

Stability and Damping: The primary objective is to maintain robust closed-loop stability as the rocket's mass $m(t)$ decreases. The underlying plant has an effective damping term

$a(t) = \frac{k - \dot{m}_f}{m(t)}$, which can become small or even unfavorable as fuel is depleted. The control law must ensure that, despite this time variation, all closed-loop poles remain in the left half-plane and the altitude dynamics stay well damped.

Trajectory Tracking: A second objective is accurate tracking of a prescribed vertical altitude reference $y_{ref}(t)$ with negligible steady-state error. As the rocket becomes lighter and the plant gain $1/m(t)$ increases, the controller must adjust its action so that rise time, overshoot, and settling time remain within acceptable bounds across the entire mass range.

Jerk Regulation: A third objective is to limit jerk $j(t) = \ddot{y}(t)$ to protect structural integrity and avoid high-frequency transients. Since differentiating the plant dynamics shows that jerk is directly influenced by the rate of change of the thrust command $\dot{u}(t)$, the control system must enforce bounds on this rate so that $|j(t)|$ stays within a specified safe range (for example, $\pm 0.5 \text{ m/s}^3$).

System block diagram

Mathematical Equations: The non-linear force balance for vertical flight, incorporating variable mass $m(t)$, gravitational acceleration g , and aerodynamic drag $D(v)$, is given by:

$$\frac{d}{dt}(m(t)v(t)) = T(t) - m(t)g - D(v(t)) \quad \dot{v}(t) = \frac{T(t)}{m(t)} - g - \frac{D(v(t))}{m(t)} - \frac{\dot{m}(t)}{m(t)}v(t)$$

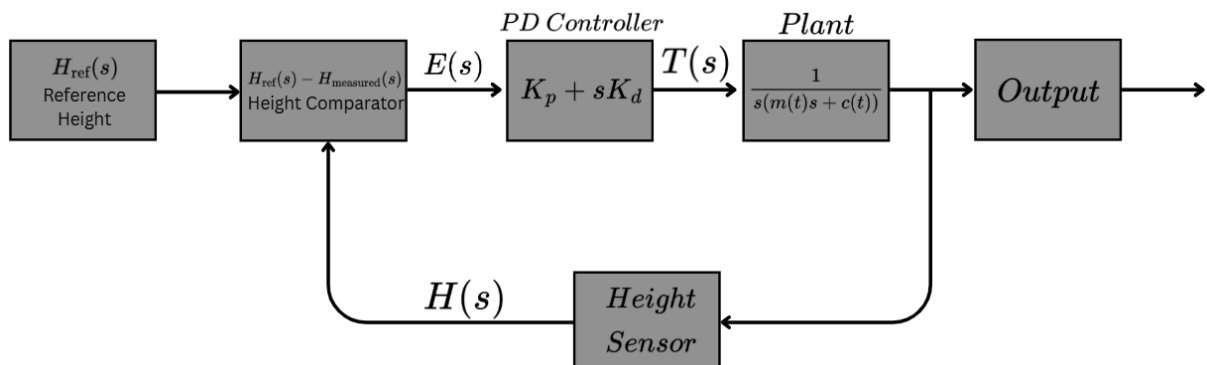
We use a linear drag model for analytical tractability: $D(v) = kv$

Assume fuel burns at constant rate: $m(t) = m_0 - \dot{m}_f t$

Since the system is time-varying, we approximate it as locally LTI by freezing mass at m^* :

$$a = \frac{k}{m^*} - \frac{\dot{m}_f}{m^*}. \quad \text{Hence, final LTI mode (Eq 1): } \ddot{y} + a\dot{y} = \frac{1}{m^*}u(t)$$

$$\text{Jerk is defined as: } j(t) = \ddot{y}(t) \quad \ddot{y} + a\dot{y} = \frac{1}{m^*}\dot{u}(t) \quad j(t) = -a\dot{y}(t) + \frac{1}{m^*}\dot{u}(t)$$



❖ **Plant Transfer Function**

Taking Laplace transform: $s^2Y(s) + asY(s) = \frac{1}{m^*}U(s)$

Thus: $P(s) = \frac{Y(s)}{U(s)} = \frac{1/m^*}{s(s+a)}$. Replacing: $a = \frac{c}{m^*}$, we obtain: $P(s) = \frac{1}{m^*s^2 + cs}$

From the Plant transfer function mentioned below, we obtain:

1. *Zero Locations*: The system has no finite zeros.
2. *Pole Locations*: The characteristic equation yields two poles: $s_1 = 0$, $s_2 = -a$

Stability Condition: $a > 0 \Rightarrow k > \dot{m}_f$

If $k > \dot{m}_f$: stable

If $k < \dot{m}_f$: unstable (negative damping dominates)

❖ **Type of controller used (P / PI / PID)**

A PD controller with Lead Compensation:

Why PD ? Because we already have a $1/s$ term in the denominator that acts as an integrator.

What does a lead compensator do?

$$C(s) = K \frac{s+z}{s+p}, \quad (z < p)$$

→ Adds a **zero** at $s = -z$

→ Adds a **pole** at $s = -p$ (further left), but suppose $p \gg z$, then $s+p \sim p$, therefore:
 $C(s) \approx K'(s+z) \approx Kp + sKd \rightarrow$ we'll be using this.

❖ **How do you make it stable**

For a system to be stable, all poles must lie strictly in the Left Half Plane (LHP).

The uncompensated system has one pole at the origin ($s=0$), so it is only marginally stable. A **lead compensator** shifts the root locus toward the Left Half Plane by adding a zero and a pole, which improves damping and moves system poles away from the imaginary axis.

The **PD controller** further improves system response by increasing stability and reducing oscillations.

❖ Performance Criteria

The characteristic equation for the close loop transfer($1 + G(s)C(s)$) function thus

becomes Closed loop characteristic equation $= 1 + \frac{K_p + sK_d}{s(ms + c)} = 0$

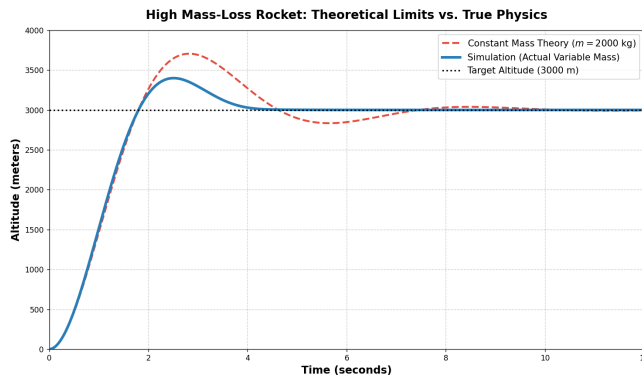
therefore : $ms^2 + (c + K_d)s + K_p = 0$, comparing with the standard equation for a

second order system ($s^2 + 2\zeta\omega_n s + \omega_n^2$) we get $2\zeta\omega_n = \frac{c + K_d}{m}$ and $\omega_n^2 = \frac{K_p}{m}$.

We can use these equations to further find the rise time($t_r \approx \frac{1.8}{\omega_n}$),

peak time($t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$), settling time($t_s \approx \frac{4}{\zeta\omega_n}$) and max peak overshoot

($M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$) and the stability criterion becomes $c + K_d > 0$ and $K_p > 0$.



Parameter	Constant Mass Theoretical values	Varying Mass Theoretical values	From Graph/Simulation
Rise Time (t_r)	1.80 s	1.68 s	~1.71 s
Peak Time (t_p)	2.82 s	2.54 s	~2.51 s
Settling Time (t_s)	6.86 s	4.39 s	~4.19 s
Max Peak Overshoot (M_p)	12.30%	17.60%	~13.3 %

❖ Tools used:

BIBO Stability: The closed-loop system achieves BIBO stability as long as $c + K_d > 0$ and $K_p > 0$.

Root Locus Method: This tool is used to analyze the movement of system poles and design the PD compensator. By adding a zero, the controller shifts the root locus toward the Left Half Plane (LHP), which improves damping and prevents the poles from drifting toward the unstable region as mass decreases.

Routh's Stability Criteria: This mathematical test is applied to the characteristic equation $ms^2 + (c + K_d)s + K_p = 0$. It confirms that the system is stable if all coefficients are positive, ensuring the rocket remains upright and controllable despite time-varying dynamics.

Steady State Error: $e_{ss} = \lim_{s \rightarrow 0} \frac{s.R(s)}{1 + G(s)}$ where $R(s) = \frac{1}{s}$, i.e. unit step input, since we want our object to go to a certain height and stay there. Therefore,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + G(s)} \quad G(s) = \frac{1}{s(ms + c)} \quad e_{ss} = 0$$

List of components you would use in this experiment

1. Microcontroller / Flight Controller
Pixhawk or Arduino for real-time control loop implementation.
2. IMU Sensor
To measure acceleration and estimate jerk.
3. Barometric Pressure Sensor
For altitude feedback.
4. Ultrasonic / LiDAR Altitude Sensor
For low-altitude accurate height measurement. +
5. ESC
To control motor speed and modulate thrust.
6. Brushless DC Motor
To generate lift/thrust.
7. Propeller / Rotor
Converts motor rotation into thrust.
8. Battery / Power Supply
Provides electrical power to the complete system.
9. Voltage Regulator / BEC
Supplies stable voltage to sensors and controller.
10. Load Cell / Thrust Sensor
To measure generated thrust experimentally.
11. Fuel/Mass Simulation Setup
A variable payload or controlled mass-reduction mechanism to represent fuel burning.
12. GPS Module
Optional, for outdoor position and altitude reference.
13. Data Logger / SD Card Module
To store altitude, acceleration, jerk, thrust, and control input data.
14. Computer / Laptop
For MATLAB/Simulink analysis, Laplace-domain simulation, and controller tuning.
15. Communication Module
Telemetry, Wi-Fi, or Bluetooth for real-time monitoring.
16. Frame / Test Rig
Mechanical structure to safely mount the motor, sensors, and controller.
17. Safety Switch / Kill Switch
To immediately stop the system during testing.
18. Wires, Connectors, and Breadboard/PCB
For electrical connections and circuit assembly.
19. Power Distribution Board
To distribute battery power to motor, ESC, sensors, and controller.
20. Software Tools
MATLAB, Simulink, Arduino IDE, or Mission Planner for modeling and testing.

Extra Information

➤ Real-World Utility

- ❖ **Managing Jerk in High-Stakes Landings:** It is used to manage the rate of change of acceleration (jerk) during critical descent phases where mass is rapidly reducing.
- ❖ **Smooth Motion Profiles:** It is used to create comfortable acceleration and deceleration profiles, such as those needed for smooth travel in vertical transport systems.
- ❖ **High-Precision Vertical Tracking:** It helps aerospace vehicles maintain a pre-defined altitude trajectory even as their physical dynamics shift due to fuel consumption.

➤ Technical Benefits and Problem Solving

- ❖ **Structural Integrity:** By bounding jerk within specific limits (e.g., $\pm 0.5 \sim \text{m/s}^3$), the system prevents mechanical stress and ensures structural safety.
- ❖ **Stability of Variable Systems:** It provides a framework to maintain stability in "variable-mass" systems where fixed settings would otherwise fail as the vehicle becomes lighter.
- ❖ **Oscillation Reduction:** The use of lead compensation and PD control helps reduce high-frequency oscillations and improves damping, ensuring the vehicle doesn't "over-correct" its position.
- ❖ **Phase Margin Maintenance:** It ensures the system has an adequate safety buffer (phase margin) throughout the entire flight, preventing the vehicle from becoming unstable during rapid maneuvers.

➤ Limitations and Challenges

- ❖ **Gain Instability:** A primary limitation is that fixed control gains become unstable as mass decreases, necessitating more advanced strategies like gain scheduling.
- ❖ **Plant Complexity:** The physical system is naturally "marginally stable" (due to a pole at the origin), meaning it requires active, constant correction just to stay upright.
- ❖ **Sensor Reliance:** The system is heavily dependent on the accuracy of real-time feedback from instruments like barometric sensors and IMUs to calculate jerk and altitude.